

SHUNYA

Final Implementation Plan

2026-07-14

9 corrections applied. 18 phases. Ready for implementation.

Normalized Phase Order (Correction 1)

- 0 – Architecture Convergence
- 1 – Unified Human Identity
- 2 – Relationship Intelligence
- 3 – Conversation & Interaction
- 4 – Privacy, Sensitivity & Memory Eligibility ← before Human Context
- 5 – Human Context ← after Privacy
- 6 – Memory Architecture ← after Privacy
- 7 – Evidence, Provenance & Source Intelligence
- 8 – Evidence Runtime Distinction
- 9 – LLM Intelligence Runtime
- 10 – Context Fusion Engine
- 11 – Internal-First Knowledge Resolution
- 12 – World Intelligence
- 13 – Relevance Intelligence
- 14 – Persistent Plans & Governed Action
- 15 – Closed Learning Loop
- 16 – Relationship-Aware Assistant
- 17 – Persistent Application Shell
- 18 – Mobile Application Runtime

Key change: Memory Architecture moves to Phase 6 (after Privacy at 4, after Human Context at 5). Memory may be designed during earlier phases but durable Human Context persistence and sensitive memory runtime activation are gated behind Phase 4 policy enforcement.

Phase 4 — Sensitivity Classification (Corrected PUBLIC semantics)

Correction 2 applied.

SENSITIVITY ≠ VISIBILITY ≠ TENANT ACCESS ≠ PURPOSE ELIGIBILITY

- PUBLIC → No sensitivity-specific restriction, but remains subject to tenant scope, visibility, purpose, provenance, context and access policy. NO cross-tenant exposure merely because data is classified PUBLIC.
- ORGANIZATIONAL → Organization-owned information; visible within tenant.
- PERSONAL → Human-personal context voluntarily shared; structurally distinct from organization-owned information.
- SENSITIVE → Requires explicit authorization.
- HIGHLY_RESTRICTED → Limited to specific authorized roles; audit-logged.

Phase 5 — Human Context (Corrected CustomerProfile semantics)

Correction 3 applied.

CustomerProfile owns: - lifetime_value (calculated from transactions) - segment (assigned or derived with provenance) - preferred_channel (EXPLICITLY selected or operationally established WITH provenance)

CustomerProfile does NOT own as a static field: - communication_style (inferred → belongs in HumanSignal/PatternCandidate/Human Preference Memory)

The runtime flow for inferred attributes: 1. Observation in conversation → HumanSignal(signal_type=communication_observation) 2. Repeated observations → PatternCandidate 3. Sufficient evidence + confidence → Human Preference Memory(preference_type=communication_style) 4. CustomerProfile resolves an ELIGIBLE PREFERENCE VIEW from Human Preference Memory (not a permanent static field)

All Prior Corrections (Accepted)

- 1. Memory Architecture with 7 memory classes, universal metadata
- 2. Interaction/Message model with channel mapping
- 3. 5-level sensitivity with PERSONAL restored
- 4. Person model with canonical_name, preferred_name, role profiles
- 5. Relationship with EXPLICIT vs INFERRED, confidence, provenance
- 6. Rollback: CODE vs SCHEMA vs DATA; additive-only; dual-read adapters

Final Phase Map (18 phases)

- 0 Architecture Convergence
 - commit baseline, add characterization tests, converge two codebases
- 1 Unified Human Identity
 - Person, PersonIdentity, IdentityResolver, Role Profiles
- 2 Relationship Intelligence
 - Relationship with EXPLICIT/INFERRED, confidence, provenance
- 3 Conversation & Interaction
 - Conversation, Interaction/Message, ConversationContext, channel mapping
- 4 Privacy, Sensitivity & Memory Eligibility
 - 5-level sensitivity, consent, retention, memory eligibility gates
 - MUST be complete before Phase 5 & Phase 6 activate

- 5 Human Context
 - HumanSignal, PatternCandidate, CustomerProfile (no static inferred fields)
 - Communication style → Human Preference Memory, not static CustomerProfile
- 6 Memory Architecture
 - 7 memory classes: Working, Conversation, Relationship, Organizational, Human Preference, Episodic, World Intelligence
 - Universal metadata: provenance, temporal validity, confidence, sensitivity, retention, visibility, memory class, supersession
 - Runtime activation gated by Phase 4
- 7 Evidence, Provenance & Source Intelligence
 - Source, Evidence, Fact, Inference models with provenance chain
- 8 Evidence Runtime Distinction
 - "Per company data" vs "per external info" vs "analysis" vs "recommendation"
- 9 LLM Intelligence Runtime
 - Provider-agnostic ModelProvider, ToolRegistry, multi-step tool use
- 10 Context Fusion Engine
 - 13 context inputs → ReasoningContext; inserted before ReasoningLayer
- 11 Internal-First Knowledge Resolution
 - KnowledgeSufficiencyEvaluator, FreshnessRequirementEvaluator
- 12 World Intelligence
 - External source adapters (news, RSS, APIs)
- 13 Relevance Intelligence
 - Organization/person/role/urgency relevance evaluation
- 14 Persistent Plans & Governed Action
 - Plan, PlanStep, PlanApproval; AI proposes, governance disposes
- 15 Closed Learning Loop
 - Candidate states OBSERVED→REPEATED→VALIDATED→PROMOTED→knowledge update
- 16 Relationship-Aware Assistant
 - Evolved CompanionEngine, CoachEngine with relationship + human context
- 17 Persistent Application Shell
 - HTMX workspace-fragment responses, Alpine stores, no full document reload
- 18 Mobile Application Runtime
 - Portrait-first, stable shell, bottom nav, no sidebar bleed